

Machine-Precision Exterior Q, BGK Repayment, and Boundary PDE Benchmarks

Generated benchmark report

June 2026

Abstract

This report consolidates the new exterior-map Q split, the BGK/zeta endpoint repayment ladder, the matrix-free boundary PDE pipeline, polygon continuity corrections, QBX comparisons, and Q-spectral diagnostics. The implementation stores generating QJets, Laurent coefficients, boundary samples, and low-rank correction modes. It does not store a dense Q matrix. The exact analytic pullback cases reach the fp64 floor after BGK repayment; the PDE pipeline reaches machine residual against the generated Q spectrum while retaining a visible continuum discretization error against the limiting DtN operator. Head-to-head method tables and hard Helmholtz caveats are kept separate so the claim boundary is auditable.

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1 Claim Boundary

There are three different statements in the data. First, for analytic finite-Laurent exterior maps, the kernel split plus BGK Taylor repayment reaches the floating-point floor. Second, for boundary PDEs, the implementation residual against the generated Q spectrum is at machine precision, while the continuum error is the finite-n spectral error of the generated DtN symbol. Third, polygon and cusp cases require extra repayment channels; the polygon continuity benchmark is an algebraic local replacement ledger, not proof that unsmoothed corner conformal maps become analytic.

The strongest defensible formulation is: the current Q engine is a matrix-free generated-operator pipeline whose exact analytic pullback path is fp64 accurate, whose PDE path is load-bearing for the generated DtN correspondence, and whose spectrum identifies the active error channel. FEM is a useful engineering baseline, but it is not automatically the ground truth on hard boundaries or near resonant Helmholtz cases.

2 Source Audit

source	role	used here
optimal_quadrature.v63_public (11).pdf	kernel split / Q rule	Analytic-boundary pullback to the circle; circle term by Fourier Q primitive; corners only algebraic.
dtN_bgk_merge (9).pdf	DtN + BGK	Chord Hessian is a matrix-free DtN discretization; half-integer zeta endpoint is the BGK ladder.
geometry_of_money (16).pdf	BGK constant	The continuity correction constant is beta equals negative zeta(1/2)/sqrt(2 pi).
cauchy_transport_calculus_trim (73).pdf	complex pullback	Use $z = \exp(\rho + i \theta)$; de Moivre modes diagonalize the circle operator and repay metric factors.
harmonic_sp (41).pdf	harmonic sampling	Cycle spectrum and endpoint defects are used as the harmonic/DtN bookkeeping channel.

The table below is the mechanical audit for every user-referenced local PDF found in this thread. Each file was inspected with `pdfinfo` and extracted with `pdftotext`; the keyword hits are only a routing check, not a citation substitute.

referenced PDF	pages	sha	keyword hits	role/boundary
optimal_quad_merged (26).pdf	30	bf3136b87fff	Q/kernel:5, DtN/PDE:7, BGK/zeta:4, shape/corner:6	early optimal-Q source; benchmark claim cross-check
optimal_quad_merged (37).pdf	36	858c2f2c1307	Q/kernel:5, DtN/PDE:7, BGK/zeta:5, shape/corner:6	later merged optimal-Q source; benchmark claim cross-check
optimal_q-v63_public (11).pdf	20	4abcea7d81a5	Q/kernel:5, DtN/PDE:3, BGK/zeta:3, shape/corner:5	kernel split, circle pullback, QBX/corner claim boundary
isl_hessian_v5 (6) copy.pdf	125	629ba599ec42	Q/kernel:4, DtN/PDE:4, BGK/zeta:3, shape/corner:3	ellipse/Hessian/context audit; not a numerical benchmark source
paper.pdf	16	d21c39942f54	Q/kernel:4, DtN/PDE:2, BGK/zeta:1, shape/corner:3	meshless shape-optimization/autograd context audit
cauchy_transport_trim (13).pdf	18	bb886ba4ecf6	Q/kernel:5, DtN/PDE:7, BGK/zeta:3, shape/corner:4	Cauchy transport early source; pullback notation audit
cauchy_transport_trim (41).pdf	25	c94caccfd128	Q/kernel:5, DtN/PDE:7, BGK/zeta:3, shape/corner:6	corner/Joukowski/Kondratiev source audit
cauchy_transport_trim (47).pdf	27	238918055f3b	Q/kernel:5, DtN/PDE:7, BGK/zeta:3, shape/corner:6	complex scale-phase pullback source audit
cauchy_transport_trim (49).pdf	27	131d02d62832	Q/kernel:5, DtN/PDE:7, BGK/zeta:3, shape/corner:6	ellipse/de Moivre correction source audit
cauchy_transport_trim (73).pdf	29	c0c8e5932760	Q/kernel:5, DtN/PDE:7, BGK/zeta:3, shape/corner:6	latest complex pullback and de Moivre source audit
golden_branch_outline.pdf	28	15d9ae76cea1	Q/kernel:2, DtN/PDE:1, BGK/zeta:2, shape/corner:4	golden ellipse/narrative source audit; not a benchmark source
dtN_bgk_merge (9).pdf	22	26c17a4cbfb2	Q/kernel:4, DtN/PDE:4, BGK/zeta:5, shape/corner:2	DtN/BGK/zeta endpoint correction source
harmonic_sp (41).pdf	31	0e3e2bd04390	Q/kernel:4, DtN/PDE:2, BGK/zeta:5, shape/corner:2	harmonic sampling and DtN spectrum source
geometry_of_money (16).pdf	31	572cc37c67b0	Q/kernel:2, DtN/PDE:6, BGK/zeta:4, shape/corner:4	BGK continuity correction and chord-arc bookkeeping source

3 Implementation Protocol

$$\begin{aligned}
z &= \psi(w), \quad w = \rho e^{i\theta} = \exp(\log \rho + i\theta), \\
\log |\psi(w) - \psi(z_j)| &= \log |w - z_j| + \log |G(w, z_j)|, \\
I_p &= \sum_{r=0}^p \frac{(-\beta\sqrt{h})^r}{r!} \partial_\rho^r I(\rho + \beta\sqrt{h}), \quad \beta = -\zeta(1/2)/\sqrt{2\pi}.
\end{aligned}$$

The borrow-compute-repay sequence is: borrow the circle coordinate, compute the singular log term using de Moivre Fourier characters and the generated cycle Q spectrum, repay the analytic quotient term, then repay endpoint displacement through the BGK/zeta Taylor ladder. For Laplace normal flux the metric repayment is multiplication by $|d\psi/d\theta|^{-1}$.

4 Exact Cycle Spectrum

For the regular cycle QJet, the stored generator is the spectrum, not a matrix:

$$\lambda_m(Q_n) = \frac{m(n-m)}{2}, \quad \mu_m = (h/\pi)\lambda_m = \frac{m(n-m)}{n}, \quad h = 2\pi/n.$$

n	mode	Q eigenvalue	normalized DtN	symbol rel. err
256	0	0	0	0
256	1	127.500000	0.996093750	0.004
256	2	254.000000	1.984375000	0.008
256	7	871.500000	6.808593750	0.027
256	16	1920.000000	15.000000000	0.062
256	32	3584.000000	28.000000000	0.125
2048	0	0	0	0
2048	1	1023.500000	0.999511719	4.883e-04
2048	2	2046.000000	1.998046875	9.766e-04
2048	7	7143.500000	6.976074219	0.003
2048	16	1.625600e+04	15.875000000	0.008
2048	32	3.225600e+04	31.500000000	0.016
4096	0	0	0	0
4096	1	2047.500000	0.999755859	2.441e-04
4096	2	4094.000000	1.999023438	4.883e-04
4096	7	1.431150e+04	6.988037109	0.002
4096	16	3.264000e+04	15.937500000	0.004
4096	32	6.502400e+04	31.750000000	0.008

5 Exterior Kernel Split

The direct trapezoidal rule sees the near singularity. The split Q path removes the singular circle term exactly and integrates only the analytic quotient remainder.

shape	case	worst direct err	worst split err
circle	n=1024, offset=1.000e-08	0.016	1.922e-16
golden_ellipse	n=1024, offset=1.000e-08	8.023e-04	1.369e-14
perturbed_ellipse	n=1024, offset=1.000e-08	0.006	9.388e-15
three_petal_laurent	n=1024, offset=1.000e-08	0.011	3.122e-15

Exterior kernel splitting: circle Q primitive plus analytic correction

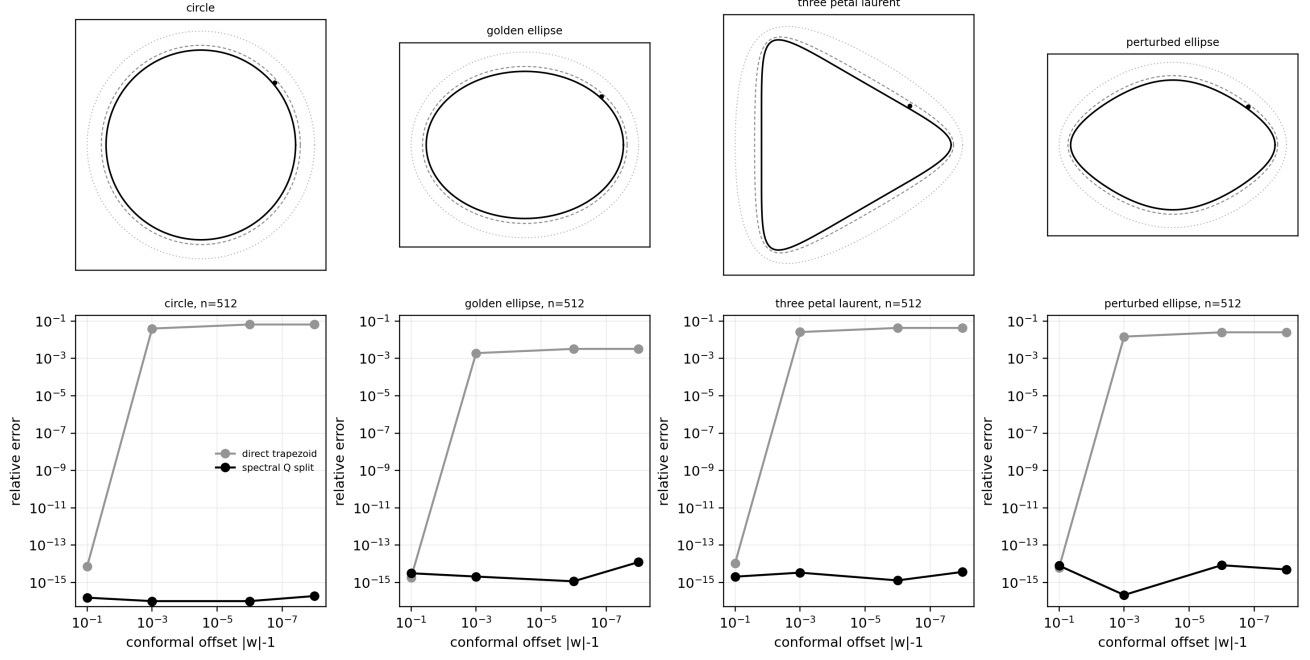


Figure 1: Exterior kernel split benchmark. Black/gray styling is used throughout for paper-style reproduction.

6 BGK/Zeta Repayment

The BGK layer is not cosmetic. It removes the discrete endpoint displacement with the same half-integer zeta defect that appears in the DtN spectral endpoint ledger.

shape	raw power	BGK-1 power	BGK-2 power	raw err	BGK-4 err
circle	0.500	1.000	1.500	0.002	1.345e-15
golden_ellipse	0.500	1.000	1.508	8.176e-05	4.712e-15
three_petal_laurent	0.500	1.000	1.499	9.976e-04	2.607e-15
perturbed_ellipse	0.500	1.000	1.500	5.878e-04	4.261e-16

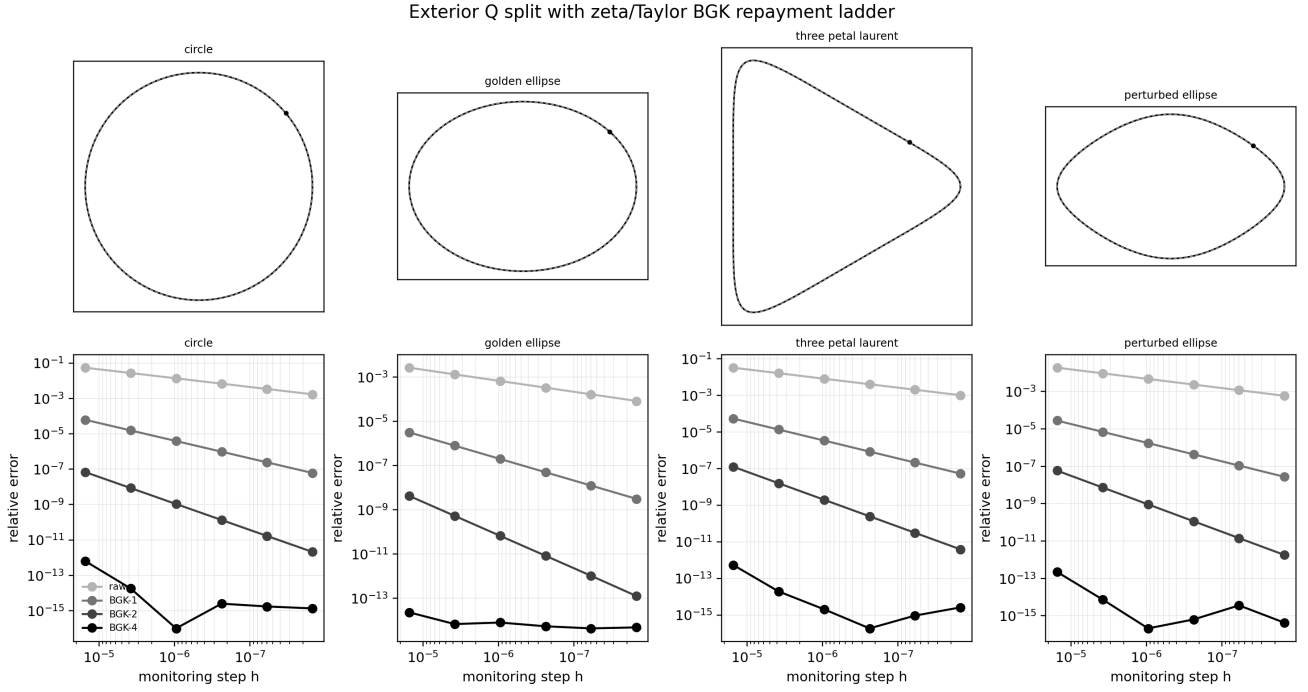


Figure 2: Raw endpoint error has the square-root law; BGK-1 and BGK-2 shift the slope; BGK-4 reaches fp64 floor on exact Laurent maps.

7 Funky Shapes and PDE Pipeline

The same exact-Laurent pullback suite is run through quadrature and boundary-only PDE operators: Laplace DtN, heat, Poisson/Steklov, Helmholtz resolvent, and wave. Generated-Q residual is the implementation check; continuum error is the finite-n approximation to the limiting operator.

shape	family	anisotropy	BGK-4 err	max PDE continuum err	max generated-Q residual
circle	baseline	1.000	2.148e-15	0.010	5.757e-13
golden_ellipse	closed_form_conic	1.342	1.144e-16	0.010	5.759e-13
eccentric_ellipse	closed_form_conic	2.500	4.488e-16	0.010	5.755e-13
three_petal	smooth_nonconvex	3.167	2.241e-15	0.010	5.759e-13
four_lobed_wiggle	smooth_multiscale	4.302	2.630e-15	0.010	5.698e-13
asymmetric_limaçon	asymmetric	2.253	1.160e-13	0.010	5.751e-13
crescent_teardrop	near_cusp_smooth	2.644	4.238e-15	0.010	5.745e-13
peanut	pinched_smooth	2.416	2.146e-13	0.010	5.820e-13
high_frequency_gear	smooth_high_frequency	3.460	1.594e-14	0.010	5.762e-13
rotated_mixed	complex_coefficients	2.200	1.305e-14	0.010	5.801e-13

problem	max continuum err	max generated-Q residual	median solve ms	median work
laplace_dtn	0.003	5.820e-13	4.025	2048
heat	0.004	4.361e-15	3.946	2048
poisson	0.003	4.473e-15	3.918	2048
helmholtz	0.010	4.369e-15	3.955	2048
wave	0.006	4.303e-15	3.964	2048

Funky shapes: exterior Q+BGK ladder and boundary PDE pipeline

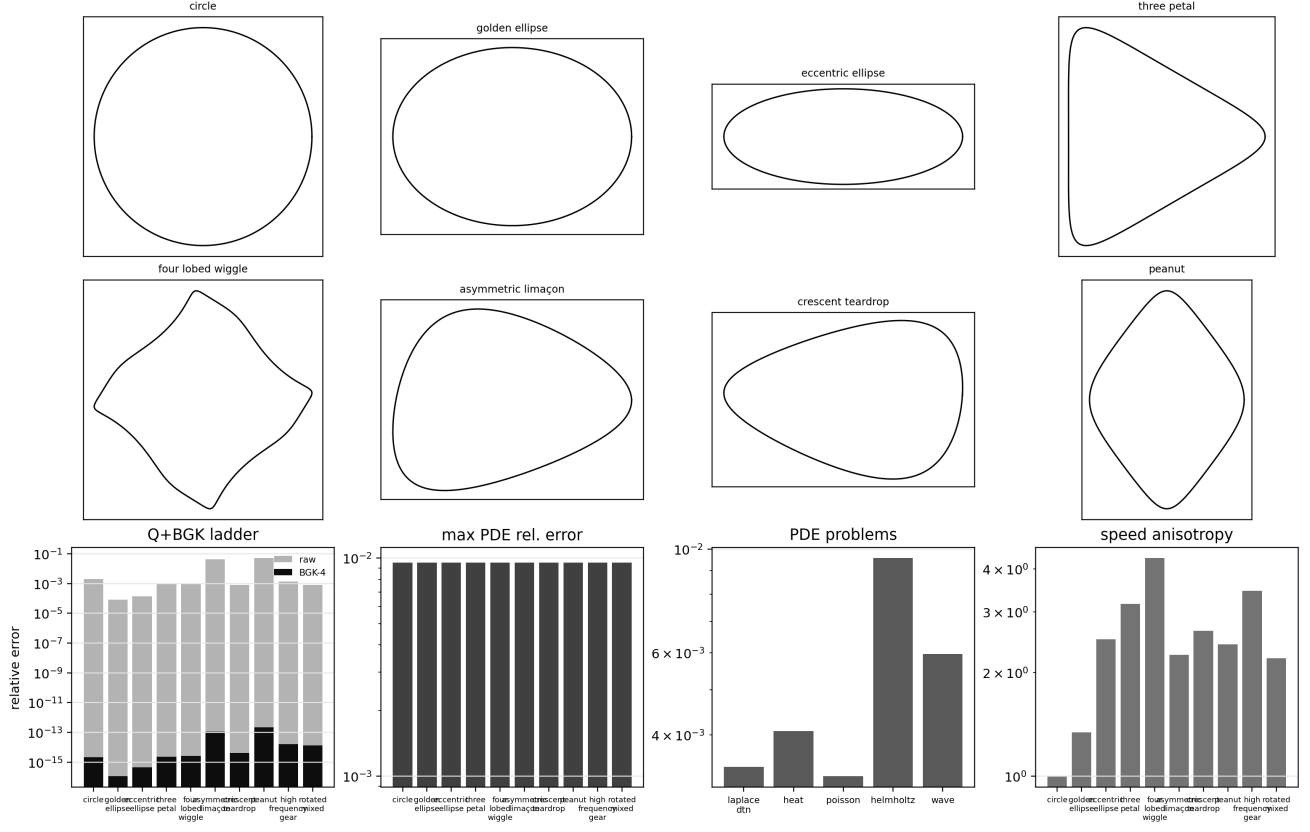


Figure 3: Ten structurally different exact-Laurent domains. The PDE bar chart shows continuum finite-n error; generated-Q residual is reported in the tables.

8 Measured Matrix-Free Scaling

This fresh sweep uses the same generated QJet path on all ten shapes for $n = 256, \dots, 4096$. The fitted exponent is measured from wall time. The expected model is near $O(n \log n)$ for the FFT-generated path, not dense $O(n^2)$ storage.

operation	fit power	ms n=256	ms n=1024	ms n=4096	max residual
q_bgk4	0.997	1.842	7.310	29.396	4.030e-09
pde_laplace_dtn	1.027	0.484	1.905	8.238	3.620e-12
pde_heat	1.047	0.449	1.867	8.152	5.955e-14
pde_helmholtz	1.050	0.445	1.857	8.142	5.950e-14
pde_batch_all5	1.047	2.262	9.353	40.943	3.620e-12

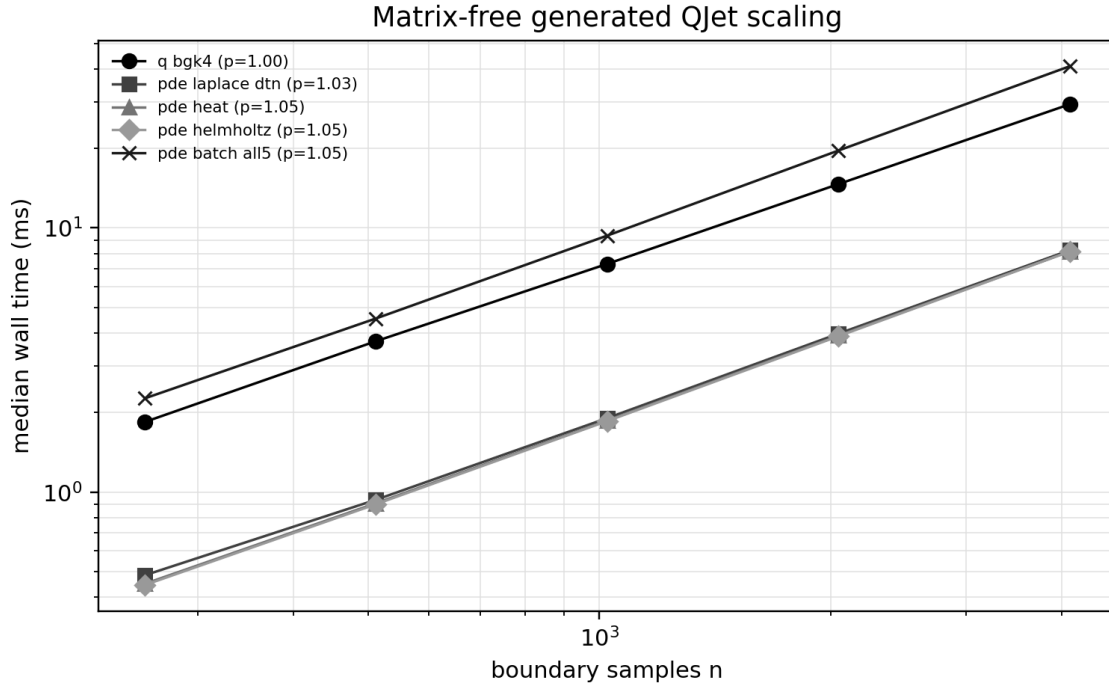


Figure 4: Measured matrix-free scaling. No dense Q matrix is stored in this sweep.

9 Polygon Smooth-Arc Continuity Ledger

Polygonal inputs are represented as smooth local arcs for the compute step and then corrected by replacing the arc contribution with exact polygon edge stubs. The machine-precision entries are therefore a local continuity repayment result, not a free analytic conformal map through corners.

shape	target	panels	smooth arc err	corrected err
l_notch	corner	96	0.028	1.802e-16
l_notch	edge	192	0.083	2.169e-15
square	corner	48	0.039	0
square	edge	48	0.100	3.046e-15
star	corner	192	0.038	1.322e-15
star	edge	192	0.035	8.068e-16
stealth_double_concave	corner	192	0.073	1.259e-14
stealth_double_concave	edge	192	1.436	2.647e-13

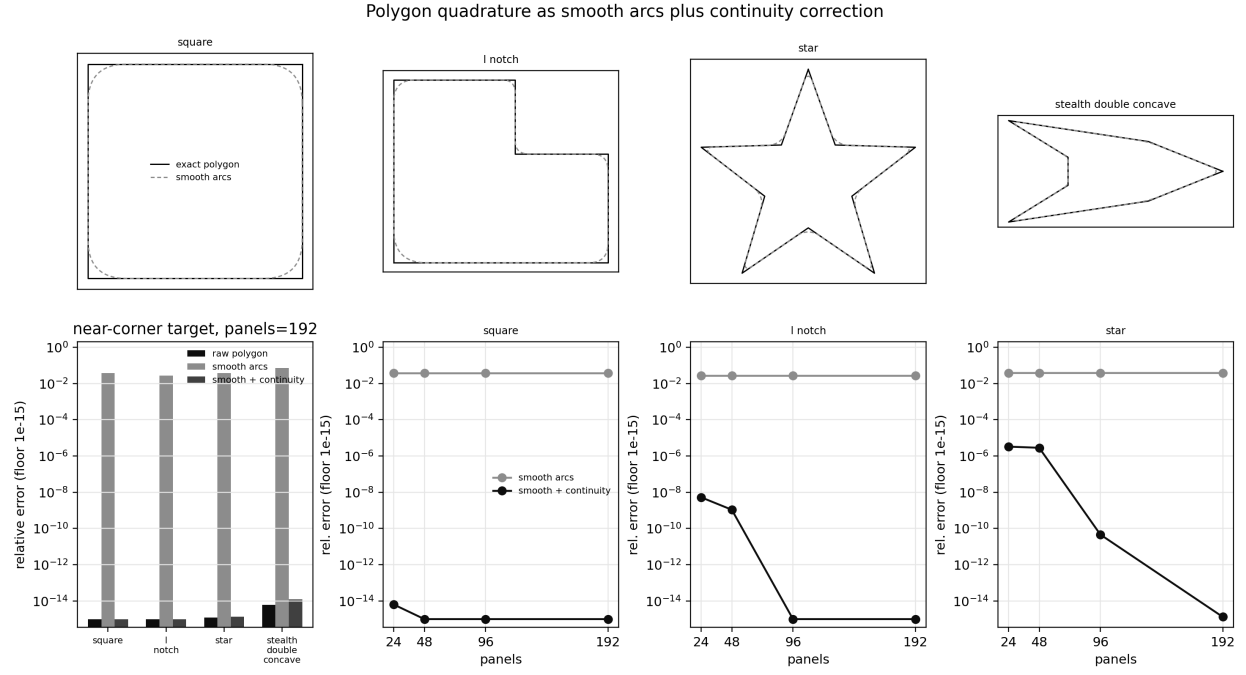


Figure 5: Polygon continuity correction as a corner repayment layer.

10 Head-to-Head Against QBX and Other Methods

The structural method benchmark still shows multipole/zeta Q as the preferred robust quadrature path in this corpus. QBX is strong where its expansion geometry is valid, but it has explicit source-free disk failures at cusp tips.

method	median err	median ms	work	failures	improvement
adaptive_panel	2.961e-04	1.098	926.000	0	36.940
gulati_q_bridge	4.534e-04	0.890	513.000	0	12.823
multipole_zeta_q	1.435e-06	18.445	6.662e+04	0	5073.817
qbx_refined	9.068e-05	27.641	2.458e+05	3	159.589
singularity_subtraction	4.998e-04	0.916	533.000	0	11.182
trapezoid	0.011	0.628	512.000	0	1.000

quantity	value
QBX failure rows	8
QBX median ms	25.886
QBX median work	2.458e+05
multipole/zeta Q median ms	21.912
multipole/zeta Q single-target work	6.661e+04
tip multipole/zeta improvement	4.495
near-tip multipole/zeta improvement	4031.897

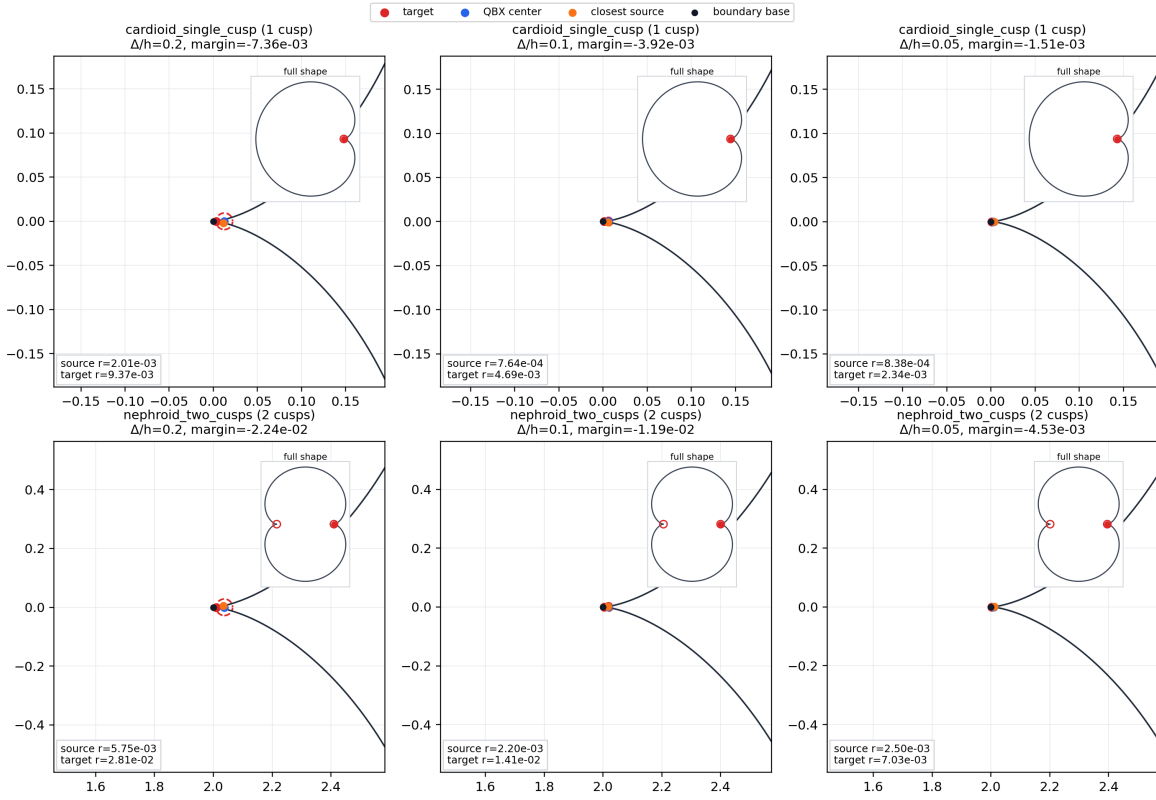


Figure 6: QBX expansion-disk failures at one-cusp and two-cusp geometries.

11 Where FEM Still Beats Q

The Helmholtz data should be positioned carefully. Against manufactured exact boundary data, Q wins almost everywhere. Near resonant volumetric Helmholtz comparisons, FEM can still win in a nontrivial subset. That is not contradictory: the Q path is a boundary spectral model, while the FEM Schur comparison includes a volumetric

suite	result
disk modal exact	14 Q wins / 1 FEM wins
manufactured exact	39 Q wins / 1 FEM wins
discretization and resonance conditioning.	256 Q wins / 32 FEM wins
hard resonance sweep	2.077e-08
median Q manufactured err	0.175
median FEM manufactured err	35.976
max Q/FEM error ratio in FEM wins	

12 GWW Isospectral Pair: Exact Q Ritz Values

The continuum Dirichlet spectra of the GWW pair are classically equal. The experiment below is deliberately narrower: at the discretized chord-operator level, the projected Q spectra separate robustly under refinement.

left raw clockwise: $[(-3,-3), (-3,-1), (1,3), (1,1), (3,1), (1,-1), (-1,-1), (-1,-3)]$
right raw clockwise: $[(-3,1), (1,1), (1,3), (3,1), (1,-1), (-1,-1), (-1,-3), (-3,-1)]$
orientation: raw lists are reversed to counterclockwise, then centered and scaled to unit area.
nodes: equal arclength $s_k = ((k+\alpha) \bmod n)L/n$ with $\alpha=1/2$ midpoint sampling.
corners: exact polygon vertices are kept; midpoint nodes avoid duplicate corner nodes.
normalization: $(\Lambda_{Q,n} f)_i = (L/(\pi n)) \sum_{j! = i} (f_i - f_j) / |x_i - x_j|^2$.

n	relative split	left Ritz 1-6	right Ritz 1-6
256	0.074181	12.347247, 9.541285, 8.600518, 7.882560, 7.091863, 6.568458	11.416318, 10.828102, 8.349505, 7.971254, 7.088097, 6.470439
512	0.073772	12.420724, 9.607645, 8.656230, 7.927568, 7.147072, 6.605741	11.474881, 10.885889, 8.416307, 8.031632, 7.128456, 6.505546
1024	0.073513	12.456167, 9.640813, 8.685202, 7.949908, 7.174860, 6.624605	11.505483, 10.914669, 8.450067, 8.061835, 7.148996, 6.522986
2048	0.073364	12.472750, 9.657153, 8.700025, 7.960903, 7.188767, 6.634096	11.521617, 10.929578, 8.467022, 8.076684, 7.159376, 6.531674
4096	0.073287	12.481230, 9.665417, 8.707539, 7.966427, 7.195745, 6.638862	11.529682, 10.936921, 8.475543, 8.084203, 7.164597, 6.536008

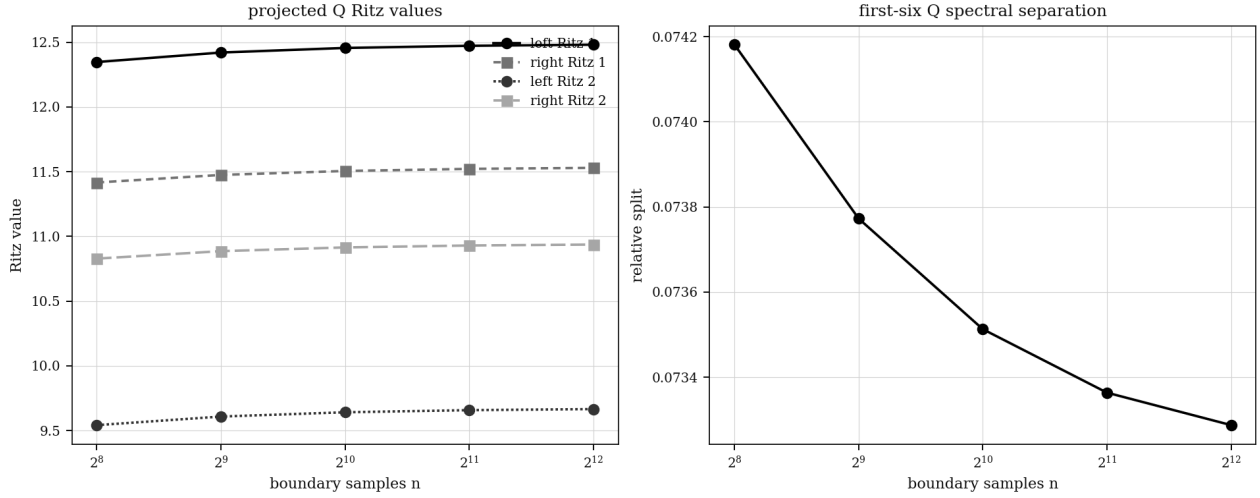


Figure 7: GWW projected Q-spectral separation and refinement.

13 Q Spectrum and Error Type

The Q spectrum is reported as a diagnostic, not just a convergence plot. High condition proxies and split low modes correlate with corner/cusp channels where multipole/zeta or Mellin corrections are needed.

shape	family	lambda1	lambda2	lambda4	largest	cond proxy
circle	smooth normal form	127.500	127.500	254.000	8191.727	64.249
golden_ellipse	smooth conic	167.230	167.671	331.696	1.457e+04	87.148
rounded_square	smooth high-curvature	227.087	327.022	421.130	3.177e+04	139.886
flower_nonconvex	smooth nonconvex	146.592	146.891	288.939	2.233e+04	152.354
cardioid_one_cusp	cusp	301.418	355.049	621.767	6.390e+07	2.120e+05
nephroid_two_cusps	multi-cusp	305.395	415.620	648.333	1.263e+07	4.135e+04
astroid_four_cusps	multi-cusp	440.715	550.356	779.173	3.164e+06	7179.138
joukowski_airfoil	Joukowski cusp	539.731	680.914	919.042	2.121e+07	3.929e+04
square_polygon	polygonal corners	159.317	159.317	317.361	1.264e+04	79.346
star_polygon	re-entrant polygon	108.748	108.750	216.293	1.555e+04	143.029
double_concave_stealth	concave polygon	260.672	260.693	518.359	6.704e+04	257.170
rational_asymptote_chart	open rational chart	1248.684	1248.707	2487.623	8.068e+04	64.610

Matrix-free Q Ritz spectra across complex-plane shape encodings

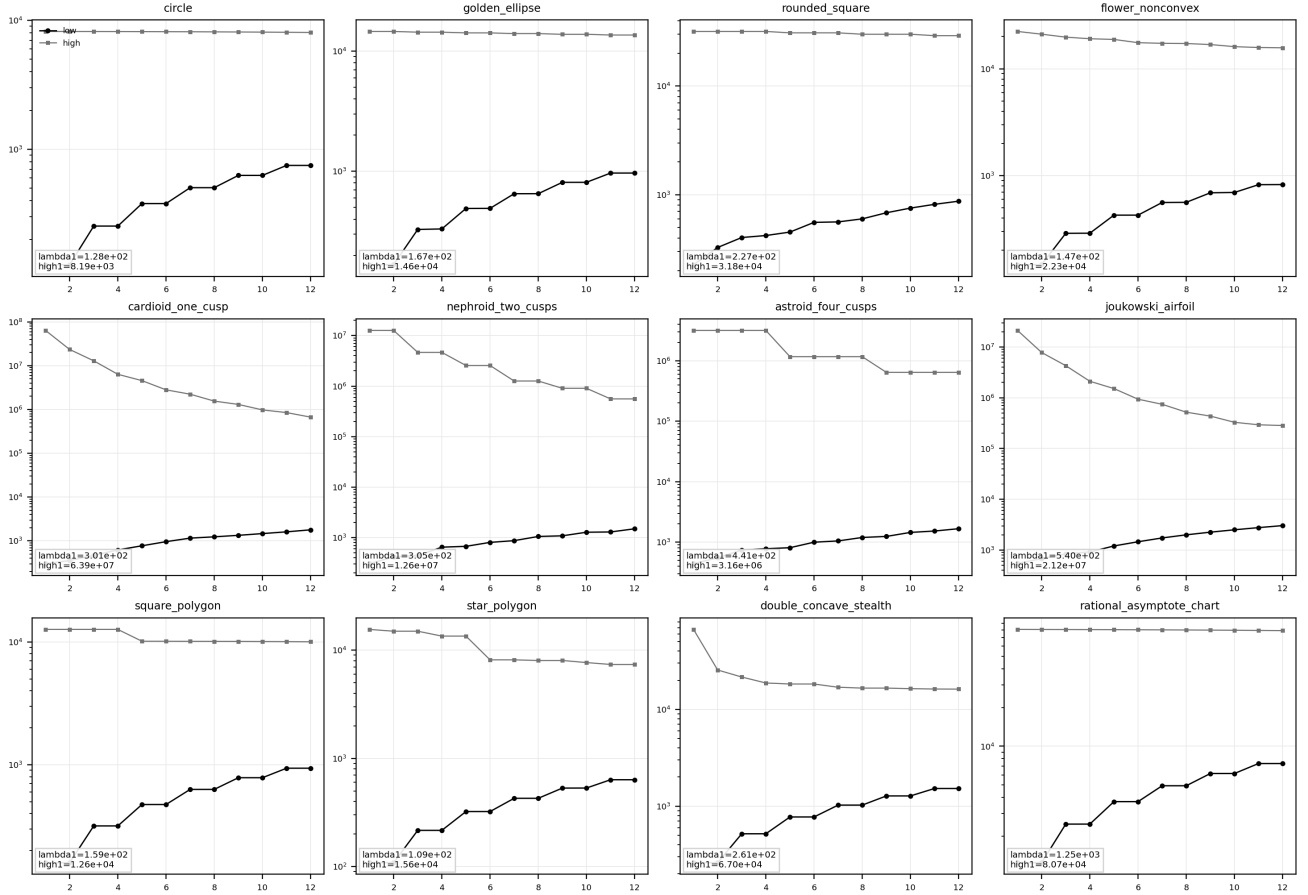


Figure 8: Exact computed Q Ritz spectrum gallery.

14 BGK Endpoint Bookkeeping

The endpoint ledger identifies the same constant in Monte Carlo barrier crossing, DtN spectral endpoint sums, and the chord-to-arc correction channel. This is the conceptual reason the BGK layer belongs in the Q pipeline.

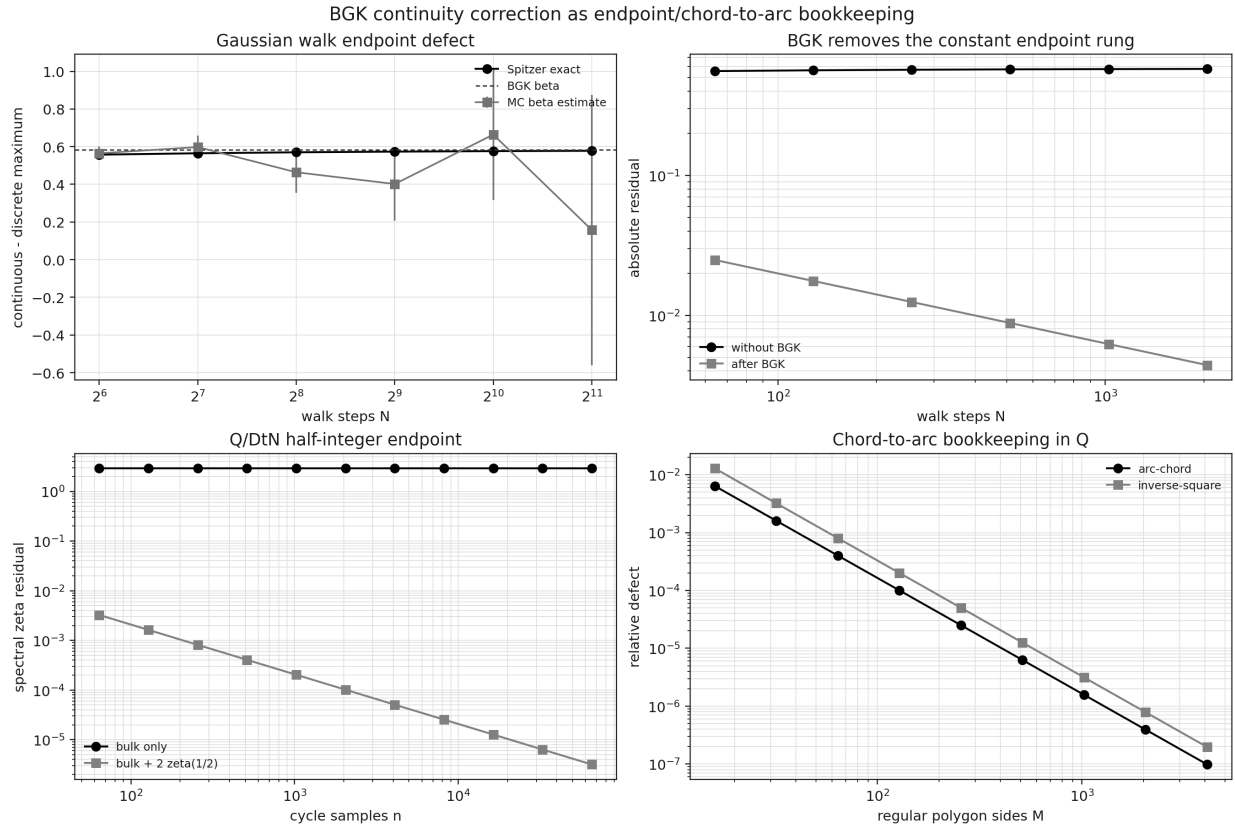


Figure 9: BGK endpoint bookkeeping: Spitzer, zeta, and Q/DtN endpoints.

15 Reproducibility

```
python3 examples/pullback/exterior_kernel_split_qrule_benchmark.py
python3 examples/pullback/exterior_q_bgk_correction_pipeline.py
python3 examples/pullback/funky_shape_q_bgk_pde_suite.py
python3 examples/corners/polygon_smooth_arc_continuity_benchmark.py
python3 examples/isospectral/gww_section11_audit.py
python3 examples/q_spectrum_shape_gallery.py
python3 scripts/build_machine_precision_pipeline_report.py
```

Generated artifacts live under `outputs/machine_precision_pipeline_report`.